Ahmedabad University
School of Engineering and Applied Science

Brain Tumor Segmentation and Localization
Using Traditional Image Processing and OpenCV

Weekly Status Report
Week 6 of 10

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1 Introduction

Brain tumor segmentation is an important step in the analysis of magnetic resonance imaging (MRI), as it helps in identifying the location and approximate boundaries of abnormal tissue. Manual segmentation can be time-consuming and depends on the experience of the person performing the segmentation. Therefore, traditional image processing methods have been studied to assist in identifying tumor regions in MRI images.

Watershed segmentation is a region-based image segmentation technique based on mathematical morphology. The basic idea is to represent a grayscale image as a topographic surface, where image intensity is treated as elevation. Low-intensity regions can be considered as valleys, and the boundaries formed when different regions meet are treated as watershed lines. This makes the method useful for finding boundaries between different regions of an image.

For brain MRI, direct application of the watershed transform can produce over-segmentation because noise, intensity variations and small image details may create several local minima. As a result, researchers have investigated multiscale and marker-controlled watershed methods to obtain more meaningful tumor regions.

The use of watershed segmentation for brain tumor MRI has been investigated directly in previous studies. Letteboer et al. evaluated an interactive multiscale watershed method for brain tumor segmentation and compared it with manual delineation in 20 patients [1]. Other studies have investigated marker-based watershed, morphological preprocessing and gradient-based approaches for improving tumor extraction [2–4].

2 Objective

The main objective of this literature review is to study the use of watershed segmentation for brain tumor segmentation and localization from MRI images using traditional image processing techniques.

The specific objectives are:

- To understand the basic principle of watershed segmentation for brain MRI images.
- To study previous applications of watershed segmentation for brain tumor extraction.
- To identify the problem of over-segmentation in the conventional watershed method.
- To study marker-controlled watershed techniques used to improve tumor segmentation.
- To study the role of preprocessing, thresholding, morphological operations and gradient information before applying watershed.
- To understand how watershed segmentation can be implemented using traditional image processing and OpenCV.

3 Literature Review

One of the direct applications of watershed segmentation to brain MRI was presented by Letteboer et al. They studied an interactive multiscale watershed algorithm for segmenting tumors in magnetic resonance brain images. The study used MRI data from 20 patients and compared watershed segmentation with manual tumor delineation. The authors reported that the watershed method produced tumor volume measurements comparable to manual delineation and was more repeatable and efficient in their evaluation [1].

The study is important for the present project because it shows that watershed segmentation can be used for actual brain tumor images rather than only for general object segmentation. However, the use of a multiscale and interactive approach also indicates that the choice of scale and image representation is important when applying watershed to medical images.

A major limitation of the conventional watershed transform is over-segmentation. Noise and small intensity variations can create additional local minima. If the flooding process starts from all these minima, the image can be divided into many small regions instead of the required tumor and background regions. Therefore, later studies have focused on controlling the watershed process through suitable markers and preprocessing.

Khan et al. proposed a brain tumor detection and classification framework based on a marker-based watershed algorithm. Their method first improved tumor contrast using gamma contrast stretching and then applied marker-based watershed segmentation for tumor extraction. The extracted regions were subsequently represented using shape, texture and point features for classification. The experiments used Harvard, BRATS 2013 and privately collected MR image datasets [2]. This study is relevant to the present project because it demonstrates how marker-based watershed can be integrated into a larger traditional image-processing pipeline.

Padlia and Sharma proposed a brain tumor segmentation method using a fractional-order Sobel mask and marker-controlled watershed transform. Regional maxima obtained from a morphologically preprocessed T2-weighted MR image were used to obtain an internal marker. A distance-transform-based watershed process was then used to obtain external marker information. A fractional Sobel mask was applied to obtain gradient information before the final watershed segmentation. The method was tested using images from the BRATS-2013 dataset [3]. This approach shows that the quality of the gradient image and marker generation can have an important effect on the final tumor boundary.

Liu et al. proposed a watershed algorithm based on morphological reconstruction and gradient layered modification for brain tumor segmentation. Their method used morphological reconstruction to smooth and denoise the original MRI image while retaining important contour information. The gradient image was then modified at different levels to remove local minima caused by irregular details and noise. Finally, morphological watershed segmentation was applied to the modified image [4]. The study is particularly relevant to the present project because it demonstrates how traditional morphological operations can be combined with watershed to reduce over-segmentation.

Hafeez, Suhail and Zwiggelaar proposed a morphological and marker-based watershed method for detection and segmentation of brain tumor regions. Their approach applied basic preprocessing and morphological operations, followed by Otsu thresholding and a marker watershed technique. The authors evaluated the method on 3000 images from the Brain Tumor Detection 2020 (Br35H) dataset and reported that appropriate preprocessing and thresholding were important for obtaining useful segmentation results [5].

More recently, Ghaderi et al. investigated marker-controlled watershed combined with fuzzy C-means clustering for automated glioblastoma segmentation from MRI images. Their method used preprocessing operations including adaptive thresholding, morphological filtering, gradient magnitude calculation and regional maxima identification before applying marker-controlled watershed. The study demonstrated that watershed can also be combined with clustering to improve the identification of tumor regions [6].

From the reviewed studies, it can be observed that watershed segmentation is generally more useful for brain tumor images when it is combined with preprocessing and marker generation rather than being directly applied to the raw MRI image. The main advantage is that watershed can provide well-defined region boundaries and can separate connected regions. Its main limitation is sensitivity to local minima, which can lead to over-segmentation. Therefore, thresholding, morphological operations, gradient processing and marker selection are important parts of a practical watershed-based brain tumor segmentation pipeline.

3.1 Comparison of Reviewed Approaches

Approach	Main Idea	Relevance to Brain Tumor Segmentation
Multiscale Watershed	Watershed segmentation is performed at different image scales	Useful for analysing tumor structures and boundaries at different scales [1]
Marker-Based Watershed	Markers are used to control the regions from which the watershed process starts	Provides better control over tumor extraction and can reduce unwanted regions [2]
Gradient-Based Watershed	Gradient information is used to emphasize possible boundaries before watershed segmentation	Useful when changes in image intensity can represent tumor boundaries [3]
Morphological Watershed	Morphological reconstruction and filtering are performed before watershed	Can reduce noise and local minima while retaining important tumor boundaries [4]
Threshold + Marker Watershed	Thresholding is used to obtain candidate regions before marker generation and watershed	Provides a relatively simple traditional image-processing pipeline for tumor segmentation [5]
Watershed + Clustering	Marker-controlled watershed is combined with fuzzy C-means clustering	Uses both clustering and watershed information to improve tumor segmentation [6]

Table 1: Comparison of watershed-based approaches reviewed for brain tumor segmentation.

4 Work Completed This Week

4.1 Methodology

Based on the literature reviewed, the general watershed-based pipeline for the project can be represented as follows:

1. Input brain MRI image is obtained and converted to a suitable grayscale representation.
2. Image preprocessing is performed to reduce noise and improve the visibility of the tumor region.
3. Thresholding or another suitable segmentation operation is used to identify possible tumor regions.
4. Morphological operations can be applied to remove small unwanted regions and improve the candidate tumor area.
5. Internal and background markers are generated.
6. The watershed transform is applied using the generated markers.
7. The watershed boundaries are used to obtain the final segmented tumor region.
8. The segmented region can then be used to determine the approximate location and boundary of the tumor.

The important observation from the literature is that watershed should not necessarily be applied directly to the raw MRI image. Preprocessing and suitable marker generation are required because unwanted local minima can result in over-segmentation.

4.2 Implementation Details

The project will use traditional image-processing operations together with OpenCV. OpenCV provides the `cv2.watershed()` function for marker-based watershed segmentation. The function uses an input image and

a marker image to perform the segmentation. The marker image contains the initial regions from which the watershed process is controlled.

For the proposed project, preprocessing can include grayscale conversion, noise reduction, contrast enhancement and thresholding. Morphological operations can then be used to improve the candidate tumor region before marker generation.

The marker-based approach is particularly relevant to this project because the main limitation of the conventional watershed method is over-segmentation. By controlling the starting regions, the number of unwanted segmented regions can be reduced.

5 Results

The literature review indicates that watershed segmentation can be useful for identifying the boundaries of tumor regions in brain MRI, particularly when it is combined with suitable preprocessing and marker generation.

The reviewed studies show that applying classical watershed directly to MRI images is generally affected by noise, intensity variations and weak or irregular tumor boundaries. Marker-controlled watershed, morphological processing and gradient-based modifications have therefore been used to improve the segmentation results.

For the proposed project, the final experimental results will be obtained after implementing the preprocessing and marker-controlled watershed pipeline on the selected MRI dataset. The segmentation results will then be compared visually and, where ground-truth masks are available, using suitable segmentation measures.

6 Challenges Faced

The main challenge identified from the literature is over-segmentation. Small intensity changes and noise in an MRI image can create additional local minima, which can cause the watershed algorithm to generate many unnecessary regions.

Another challenge is the relatively low contrast between some tumor regions and surrounding brain tissues. Tumor boundaries may not always have a strong intensity transition, which makes it difficult to generate accurate markers.

Marker selection is therefore an important part of the method. Poor markers can either cause the tumor to be divided into multiple regions or cause parts of the tumor to be missed. The reviewed literature suggests that preprocessing, thresholding, morphological operations and gradient information can be used to obtain better markers.

A further challenge is that different MRI images may require different preprocessing parameters. Therefore, parameters such as threshold levels, morphological kernel size and marker-generation criteria may need to be tested on the selected dataset.

7 Plan for Next Week

The next stage of the project will focus on implementing and testing the watershed-based brain tumor segmentation pipeline using traditional image processing and OpenCV.

The planned work is:

- Collect and organize the brain MRI images to be used for the project.
- Implement grayscale conversion and basic preprocessing.
- Test thresholding methods for identifying possible tumor regions.

- Apply morphological operations for noise removal and region refinement.
- Generate markers for marker-controlled watershed.
- Implement the watershed algorithm using OpenCV.
- Compare the segmented tumor region with the original MRI image and available ground-truth masks.
- Select suitable preprocessing and watershed parameters for the final project pipeline.

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